INTRODUCTION TO DISTRIBUTIONS

Learning objectives

- Recognise what a probability or data distribution is
- Identify when a distribution is considered to be Binomial, Poisson, or Normal
- Identify when a distribution can be treated as Normal and what this means for analytical methods

DISTRIBUTION

A **Distribution** is the location of the values in a column of data on a number line in relation to the other values – what shape does it make.

You might also hear the phrase the **structure** of the data- particularly when we begin talking about more than one column (with a slightly different meaning to *structured data*).

To see the distribution or structure of the data we don't really need to know the scale.

We can see the distribution using a **histogram** to display:

- Raw or scaled data
- Probability of each value in the dataset

NORMAL V BINOMIAL DISTRIBUTIONS

Normal Distributions

- Are seen when the data recorded or visualised in a histogram is **continuous** in nature.
- Example:
  The number of times a bullseye gets hit in a darts tournament or the recorded height of individuals in a group. (A tournament has no fixed end point)
- Scientists typically assume that a series of measurements taken from a population will be normally distributed when the sample size is large enough.

Binomial Ditributions

- Are seen when the data recorded is **discrete** in nature. The number of events (trials) is fixed and you are counting the number of successes out of those trials.
- Example
  The number of times a 6 is rolled out of 10 die rolls.

NORMAL V BINOMIAL DISTRIBUTIONS

Normal Distributions

Imagine a graph of income levels in a country. If most people are in the middle or lower income brackets, and a few people have very high incomes, the income distribution would be positively skewed.

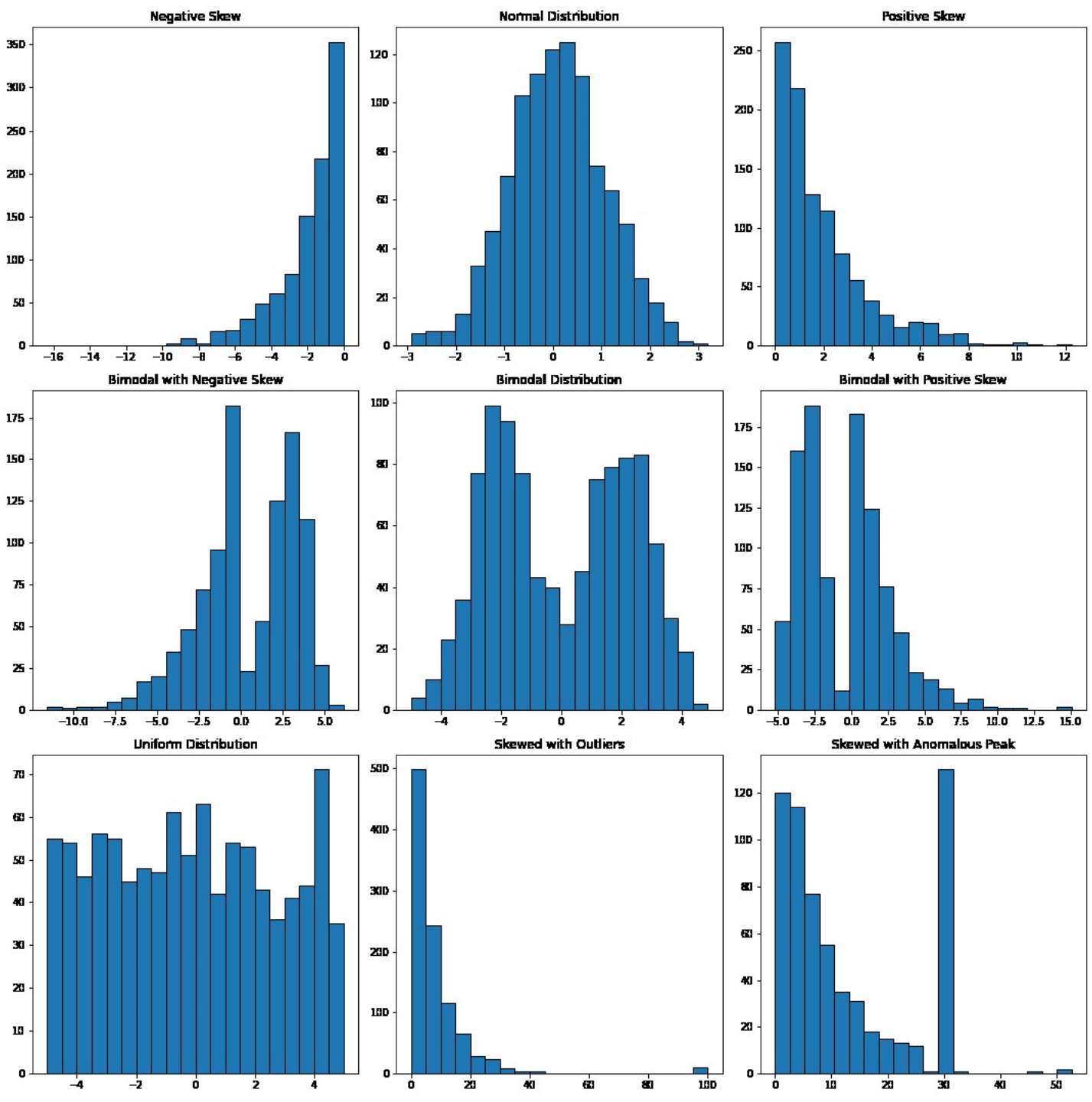
By overlaying a normal distribution curve on the histogram, you can visually compare the observed data to a theoretical normal distribution and assess how well it fits.

Binomial Ditributions

By observing the shape of the histogram, one can assess the overall likelihood of success (or failure) in the experiment.

If the histogram is skewed towards the left, it suggests that more successes are likely than failures and vice versa.

DISTRIBUTIONS: RAW DATA USING HISTOGRAMS

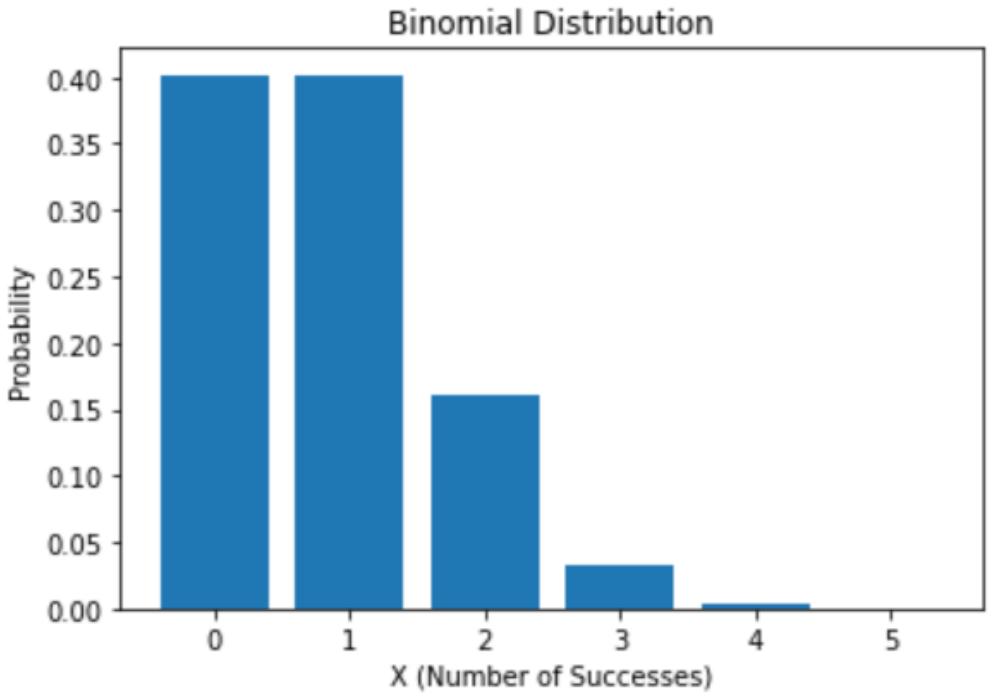


PROBABILITY DISTRIBUTION

This shows the probability of each value that the random variable X could take.

X in this case is the number of 6s rolled out of five dice.

What shape distribution do you see in this case?



BINOMIAL DISTRIBUTION

This is called a Binomial distribution because each time it's either a 6 or it's not (two choices).

X is the total number of 'successes' (getting a 6!) each time out of n rolls.

Notation

X ~ B(n, p)

This reads '**X is distributed Binomially**, with parameters n – number of trials, and p – probability of success each trial'.

In this case:

X ~ B(5, 1/6)

The values of n and p are all that's needed for the calculator to work out the distribution.

BINOMIAL USE CASES

There are 20 parts sampled from a manufacturing run – each either fails quality control or it passes. The probability is the same each time: 0.1 chance of failing.

There are 10 sets of traffic lights – you either have to stop or you don't. The probability is the same each time: 0.3 chance of stopping.

In a market research investigation, a focus group of 18 people are each independently asked if they like a product or not. The probability is 50:50 each time.

A day trader makes 15 trades a day – each is either profitable or not. The probability is the same for each one: 0.9 chance of being profitable.

BINOMIAL DISTRIBUTION: SETTING UP THE PROBLEM

There are 10 sets of traffic lights – you either have to stop or you don't. The probability is the same each time: 0.3 chance of stopping.

Define p and X in words and as a distribution

p is the probability that a randomly selected traffic light (out of all potential traffic lights) is red.

X is the number of traffic lights that are red out of 10 sets.

$X \sim B(10, 0.3)$

Binomial distribution: Python

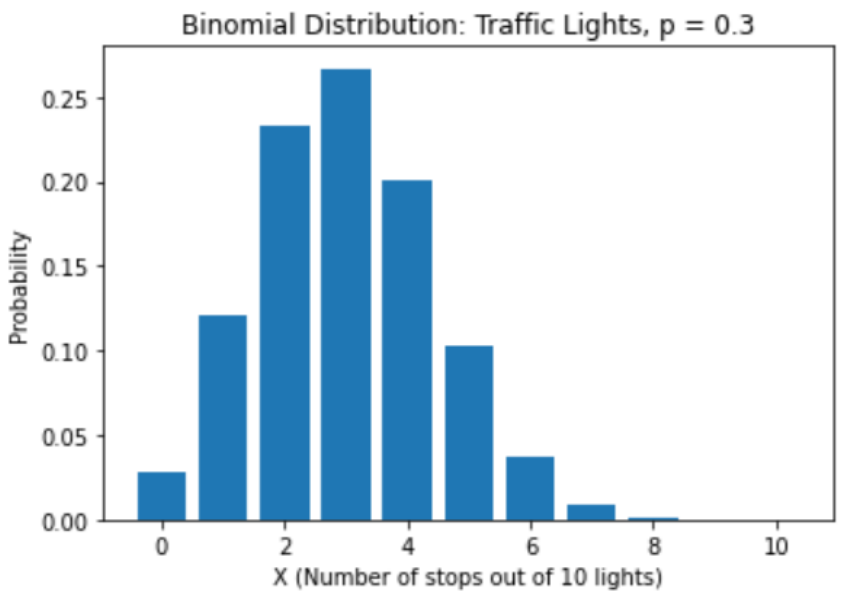
```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import binom

n = 10  # Number of trials
p = 0.3  # Probability of success each time

# Define the binomial distribution
binomial_dist = binom(n, p)

# Calculate probabilities for different values of X
x_values = np.arange(0, n+1)  # Values of X: 0, 1, 2, ..., n
probabilities = binomial_dist.pmf(x_values)

# Create a bar plot to visualize the probability distribution
plt.bar(x_values, probabilities)
plt.xlabel('X (Number of stops out of 10 lights)')
plt.ylabel('Probability')
plt.title('Binomial Distribution: Traffic Lights, p = 0.3')
plt.show()
```



ACTIVITY: BINOMIAL DISTRIBUTION

TRY THIS AND DESCRIBE THE DISTRIBUTION!

Choose one of the remaining 3 examples and **define p and X in words, and as a distribution.**

- There are 20 parts sampled from a manufacturing run – each either fails quality control or it passes. The probability is the same each time: 0.1 chance of failing.
- In a market research investigation, a focus group of 18 people are each independently asked if they like a product or not. The probability is 50:50 each time.
- A day trader makes 15 trades a day – each is either profitable or not. The probability is the same for each one: 0.9 chance of being profitable.

Extension: Write code to visualise the distribution.

SOLUTIONS

- $X \sim B(20, 0.1)$

p is the probability that a randomly selected part (out of all the possible parts) fails quality control

X is the number of parts that fail quality control out of 20.

- $X \sim B(18, 0.5)$

p is the probability that a randomly selected person (out of all the possible people) likes the product.

X is the number of people that fail like the product out of 18 people.

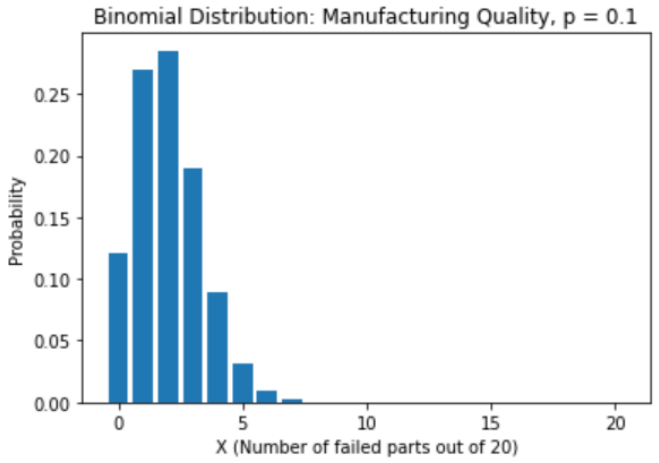
- $X \sim B(15, 0.9)$

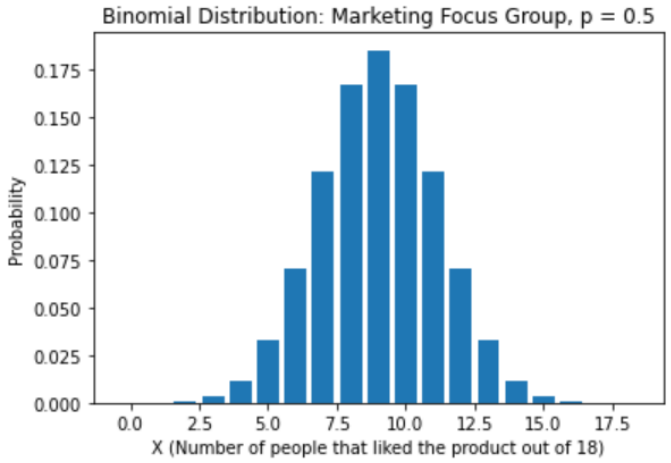
p is the probability that a randomly selected trade (out of all of the trader's possible trades) is profitable.

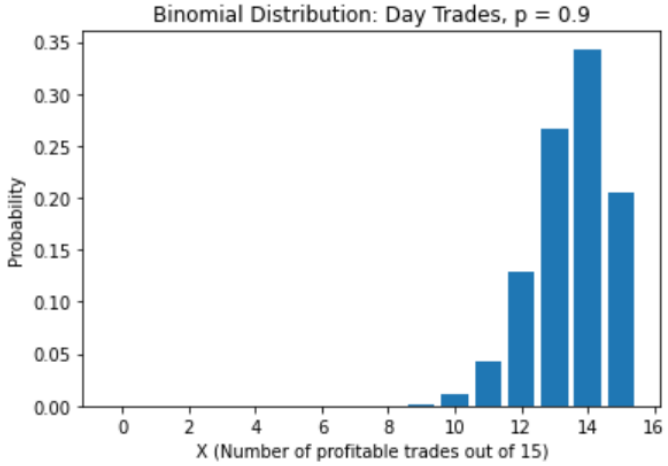
X is the number of trades that are profitable out of 15 trades.

EXTENSION SOLUTIONS

Notice the different skews for different values of p. Which looks like a **normal distribution**?







DISTRIBUTION RULES: BINOMIAL

$X \sim B(n, p)$

Parameters: n, number of trials; p, probability of success for each trial.

- Use when you have a discrete random variable representing the number of successes in a fixed number of independent trials, where each trial has two possible outcomes (success or failure).

- The trials are independent, and the probability of success remains constant across all trials.

- Examples: Coin flips, counting the number of defective items in a sample, success/failure outcomes in surveys or experiments.

DISTRIBUTION RULES: POISSON

$X \sim P(\lambda)$

Parameters: λ, “lambda” – the average rate of events occurring in a specified time or space interval.

- Use when you have a discrete random variable representing the number of events that occur in a fixed interval of time or space.

- Events occur randomly and independently, with a constant average rate.

- The events are rare in relation to the size of the interval.

- Examples: Counting the number of phone calls received in an hour, the number of emails received in a day, the number of times a car stops on a 100 mile route.

METHODS THAT RELY ON NORMAL DISTRIBUTION

There's a lot of methods that rely on the normal distribution:

- Hypothesis testing: For example, T-tests, ANOVA, F-distribution, Chi-Squared tests
- Estimating Confidence Intervals and Standard Errors
- Outlier and Anomaly detection
- Regression Analysis: Linear regression, Logistic regression

Many of these underpin more advanced methods. For example:

- Autoregressive methods for time series forecasting
- Neural Network calculations

Note that for **large sample sizes** much of this list is robust to differences from the normal distribution. This is one reason why we use very large datasets for training Machine Learning and AI models.

HOW DID YOU GET ON?

Learning objectives

- Recognise what a probability or data distribution is
- Identify when a distribution is considered to be Binomial, Poisson, or Normal
- Identify when a distribution can be treated as Normal and what this means for analytical methods